CSE 144 Final Project Report

Team

- Name(s): TODO
- Kaggle team name: TODO

Problem

This project solves the UCSC CSE 144 transfer-learning challenge. The dataset contains 100 image classes, with numeric labels matching the folder names `train/0` through `train/99`. The goal is to predict labels for the IDs listed in `sample_submission.csv`.

Method

The primary submission pipeline uses pretrained image encoders from `timm` to extract normalized embeddings. It trains a small ensemble on top of those embeddings:

- balanced multinomial logistic regression
- cosine class-prototype classifier
- cosine k-nearest-neighbor classifier

The final probabilities are blended and the best regularization / neighbor settings are selected by stratified cross-validation on the training set.

The optional fine-tuning pipeline uses a pretrained ConvNeXt model with RandAugment, label smoothing, mixup, AdamW, and cosine learning-rate decay. After validation, the model can be retrained on all labeled training images for the final checkpoint.

Reproducibility

- Fixed seed: `144`
- Device selection: CUDA, MPS, then CPU
- Label mapping: folders are sorted numerically, so folder `37` always maps to label `37`
- Submission rows: generated from `sample_submission.csv` by default

Experiments

Fill this table after running the scripts.

Run	Model / Method	Validation Accuracy	Kaggle Public Accuracy	Notes
1	ConvNeXt embedding quick run	0.7692 +/- 0.0087	TODO	One-backbone local CV; generated `submissions/submission_embedding_quick.csv`
2	ConvNeXt + Swin embedding ensemble	0.7905 +/- 0.0107	TODO	Two-backbone local CV; generated `submissions/submission_embedding_2backbone.csv`
3	ConvNeXt + Swin tuned ensemble	0.8035 +/- 0.0100	TODO	Tuned two-backbone local CV
4	ConvNeXt + Swin + DINOv2 ensemble	0.8656 +/- 0.0122	TODO	Added `vit_base_patch14_reg4_dinov2.lvd142m`
5	Block-weighted DINOv2 ensemble	0.8767 +/- 0.0151	TODO	Best local CV; feature weights `(0.25, 0.75, 1.5)`, pure logistic regression
6	ConvNeXt fine-tune	TODO	TODO	Full-train checkpoint

Final Submission

- Final CSV: TODO
- Kaggle public leaderboard score: TODO

- Model weights link: TODO
- Leaderboard screenshot: `assets/kaggle_leaderboard.png`

How to Run

See the repository README for exact setup, training, and inference commands.